

Next we sample these signals, and use the “if periodic, then out” rule from Eq. (5.19):

$$\begin{aligned} E^* &= R^* - M^* G_2^* \\ M^* &= E^* (H G_1)^* \end{aligned} \quad (5.44)$$

We solve these equations by substituting for M^* in Eq. (5.44) from Eq. (5.43)

$$\begin{aligned} E^* &= R^* - E^* (H G_1)^* G_2^* \\ &= \frac{R^*}{1 + (H G_1)^* G_2^*} \end{aligned} \quad (5.45)$$

To obtain Y we use the equation

$$\begin{aligned} Y &= E^* H \\ &= \frac{R^* H}{1 + (H G_1)^* G_2^*} \end{aligned} \quad (5.46)$$

and

$$Y^* = \frac{R^* H^*}{1 + (H G_1)^* G_2^*} \quad (5.47)$$

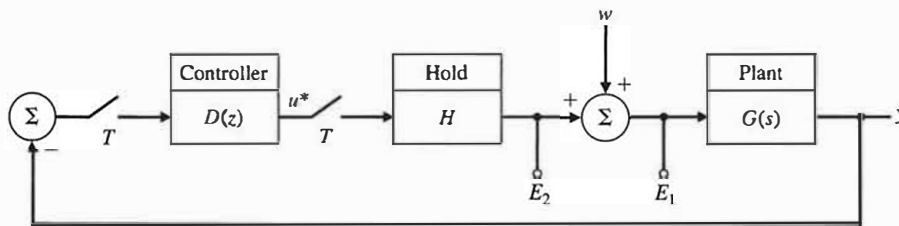
In this case we have a transfer function. Why? Because only the samples of the external input are used to cause the output. To obtain the z -transform of the samples of the output, we would let $e^{sT} = z$ in Eq. (5.47). From Eq. (5.46) we can solve for the continuous output, which consists of impulses applied to $H(s)$ in this case.

As a final example of analysis of sampled-data systems we consider a problem of experimental transfer function measurement in a sampled-data system.

◆ Example 5.5 Measuring the Transfer Function of a Sampled-Data System

It has been proposed to use an experiment to measure the loop gain of a trial sampled-data design on the actual physical system using the setup of Figure 5.16. The proposal is to have zero reference input but to inject a sinusoid into the system at W and to measure the responses

Figure 5.16
A block diagram for experimental measurement of a sampled-data transfer function



at that frequency at locations E_1 and E_2 . It is thought that the (complex) ratio of these signals will give the loop gain from which the gain and phase margins can be determined and with which a frequency response design can be worked out.

1. Compute the transforms of E_1 and E_2 for a general signal input at w .
2. Suppose that the signal w is a sinusoid of frequency ω_0 less than π/T (no aliasing). Plot the spectra of GW and $(GW)^*$ and show that $(GW)^* = \frac{1}{T} GW$ at the frequency ω_0 .
3. Use the results of 2) to get an expression for the complex ratio of the signals E_1 and E_2 when $\omega_0 < \pi/T$.
4. Repeat these calculations for the setup of Fig. 5.17 where the input signal is first sampled and held before being applied to the system.

Solution.

1. Following the procedure just given, we express the signals of interest in terms of sampled signals as follows

$$E_1 = W + U^* H \quad (5.48)$$

$$E_2 = U^* H \quad (5.49)$$

$$Y = WG + U^* HG \quad (5.50)$$

$$Y^* = (WG)^* + U^* (HG)^* \quad (5.51)$$

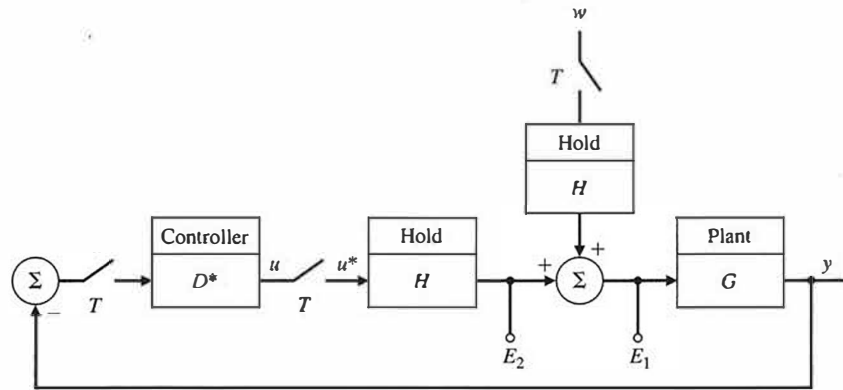
$$U^* = -D^* Y^* \quad (5.52)$$

Solving Eq. (5.52) for U^*

$$U^*(s) = -\frac{D^*(WG)^*}{1 + D^*(HG)^*} \quad (5.53)$$

Figure 5.17

A block diagram for experimental measurement of a sampled-data transfer function with sampled input



If we now substitute this result into Eq. (5.48) and Eq. (5.49) we have the solution of this part as

$$\begin{aligned} E_1 &= W - \frac{D^*(WG)^*}{1 + D^*(HG)^*} H \\ E_2 &= -\frac{D^*(WG)^*}{1 + D^*(HG)^*} H \end{aligned} \quad (5.54)$$

Clearly we do not have a transfer function since the transform of the signal is imbedded in the signal transforms.

2. For the second part, we can consider the sinusoid one exponential at a time and consider $w = 2\pi\delta(\omega - \omega_0)$. Then

$$(GW)^* = \frac{1}{T} \sum_{k=-\infty}^{k=\infty} G(j\omega - jk\frac{2\pi}{T}) 2\pi\delta(\omega - \omega_0 - \frac{2\pi k}{T}).$$

The spectra involved are easily sketched. Since $\omega_0 < \pi/T$ there is no overlap and at ω_0 the signal is

$$\begin{aligned} (GW)^* &= \frac{1}{T} G(j\omega_0) 2\pi\delta(\omega - \omega_0) \\ &= \frac{1}{T} GW|_{\omega_0}. \end{aligned} \quad (5.55)$$

3. If we substitute Eq. (5.55) into Eq. (5.54) and take the ratio, we find the describing function

$$\frac{E_2}{E_1} = -\frac{\frac{1}{T} D^*GH}{1 + D^*(GH)^* - \frac{1}{T} D^*GH} \quad (5.56)$$

Notice that if $|G| = 0$ for $|\omega| > \pi/T$ so that $G^* = G$ for frequencies less than π/T , then Eq. (5.56) reduces to

$$\frac{E_2}{E_1} = -D^*(GH)^*,$$

which is the transfer function. Thus the proposed method works well if there is a good antialias filter in the loop.

4. With the input applied through a sample and hold as drawn in Fig. 5.17 the key expressions are given by

$$\begin{aligned} E_1 &= U^*H + W^*H \\ E_2 &= U^*H \\ U^* &= -D^*Y^* \\ Y &= U^*HG + W^*HG. \end{aligned} \quad (5.57)$$

These equations can be readily solved, after taking the "star" of Y to give

$$\begin{aligned} E_1 &= \frac{W^*H}{1 + D^*(HG)^*} \\ E_2 &= -\frac{D^*(HG)^*}{1 + D^*(HG)^*} W^*H. \end{aligned}$$

From these, the ratio gives the true discrete transfer function

$$\frac{E_2}{E_1} = -D^*(HG)^*.$$

5.5 Calculating the System Output Between Samples: The Ripple

In response to a particular input, the output of a sampled-data system at sampling instants can be computed by the z -transform, even in those cases where there is no transfer function. However, in many problems it is important to examine the response between sampling instants, a response that is called the “ripple” in the response. Often, for example, the maximum overshoot will not occur at a sampling instant but at some intermediate point. In other cases, hidden oscillations are present, oscillations that may or may not decay with time. The ripple is generated by the continuous-time part of the system at the output. For example, in the case drawn in Fig. 5.13, the ripple is given by the response of $G(s)/s$ between sampling instants. Three techniques have been suggested to compute ripple. The first, suggested by J. Sklansky, is based on the partial-fraction expansion of $G(s)/s$. The second, suggested by E. Jury, is based on introducing a time shift in the sampler at the output of the system. If this shift is less than a sample period, the new samples are taken between the system samples. The modified transform from input samples to shifted samples is called the *modified z -transform* of $G(s)/s$. The third technique, introduced by G. Kranc, is based on sampling the output at a faster rate than the feedback loop is updated. Block diagrams representing the three methods are given in Fig. 5.18.

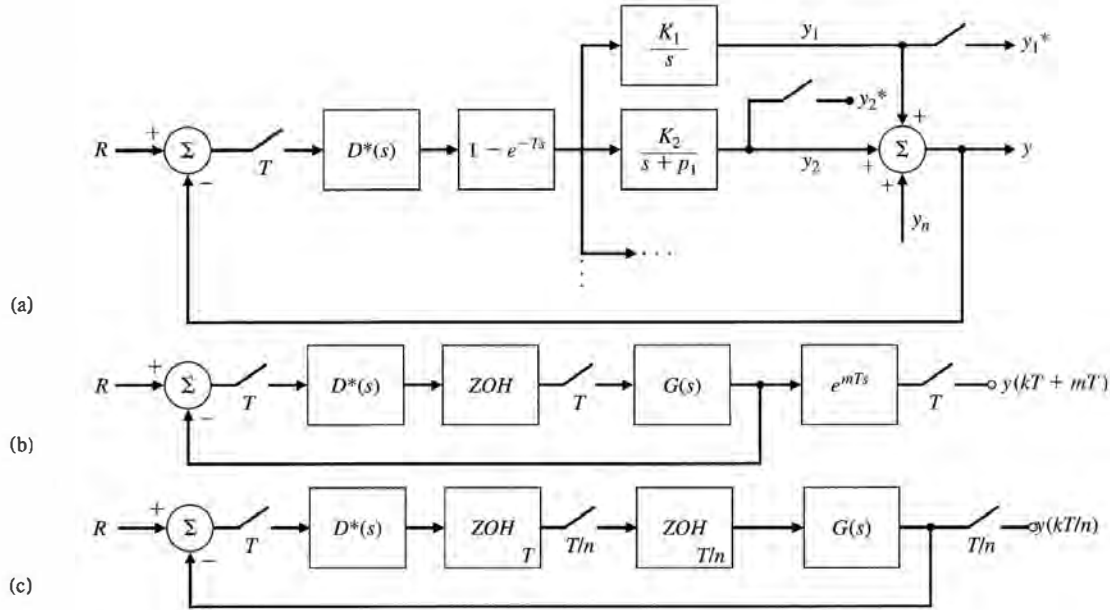
In the partial-fraction method, shown in Fig. 5.18(a), the outputs of the several fractions are sampled and the values of $y_1(kT)$, $y_2(kT)$, $\dots$ are computed in the regular way with z -transforms or MATLAB statements. These values at the instant kT represent initial conditions for the several partial fraction continuous dynamics at time kT and from them the transient over the period from kT to $(k+1)T$ can be computed. The total system output is the sum of these components. The method is somewhat tedious but gives an exact expression for the ripple during any given sample period from which, for example, the peak overshoot can be exactly computed.

The modified z -transform of the plant with zero-order hold is defined as

$$G(z, m) = (1 - z^{-1})\mathcal{Z} \left\{ \frac{G(s)}{s} e^{mTs} \right\} \quad 0 \leq m < 1,$$

Figure 5.18

Three methods used to evaluate ripple. (a) Partial fraction expansion; (b) Modified z-transform; (c) Multirate sampling



and represents samples taken at the times $kT + mT$. The modified transform of the output of the system shown in Fig. 5.18(b) is given by

$$Y(z, m) = \frac{D(z)G(z, m)}{1 + D(z)G(z)} R(z), \quad (5.58)$$

and its inverse will give samples at $kT + mT$. The modified operation is noncausal but is only being used as a computational device to obtain inter-sample ripple. For example, if $m = 0.5$ then use of Eq. (5.58) will find sample values halfway between sample updates of the control. MATLAB only permits delays (causal models) and can be used to find the output of the modified plant delayed by one sample shown in the figure as $z^{-1}Y(z, m)$. If the plant is given in state form with description matrices $[F, G, H, J]$, then the representation of the delayed modified transform can be computed in MATLAB using

$$\text{SYS} = \text{ss}(F, G, H, J).$$

The delay for sample period T and shift m is set by the command

$$\text{set}(\text{SYS}, 'td', (1 - m)T),$$

and finally, the discrete representation of the system which has a delayed modified z - transform is given by the standard conversion

$$\text{SYSD} = \text{c2d}(\text{SYS}, T).$$

The method of multi-rate sampling is shown in Fig. 5.18(c). The output of the controller is held for the full T seconds but this signal is again sampled at the rate T/n for some small n , such as 5. The plant output is also sampled at the rate T/n . The feedback loop is unchanged by these additional samplers but the output ripple is now available at n points in between the regular sample times. This technique is readily programmed in MATLAB and is regularly used in this book to compute the ripple. An interesting case is given in Fig. 7.14 where it can be seen that the maximum overshoot occurs in the ripple.

5.6 Summary

In this chapter we have considered the analysis of mixed systems that are partly discrete and partly continuous, taking the continuous point of view. These systems arise from digital control systems that include A/D and D/A converters. The important points of the chapter are

- The large-signal behavior of an A/D converter can be modeled as a linear impulse modulator followed by a zero-order-hold.
- D/A converter can be modeled as a zero-order-hold.
- The transform of a sampled signal is periodic with period $2\pi/T$ for sample period T .
- Sampling introduces aliasing, which may be interpreted in both the frequency and the time domains.
- The sampling theorem shows how a band-limited signal can be reconstructed from its samples.
- Interconnections of systems that include sampling can be analyzed by block-diagram analysis.
- If the input signal to a sampled-data system is not sampled, it is impossible to define a transfer function.
- The output of a sampled-data system between sampling instants can be computed using partial fraction expansion, using the modified z - transform, or by multi-rate sampling. With a computer, multi-rate sampling is the most practical method.

5.7 Problems

- 5.1 Sketch a signal that represents bounded hidden oscillations.

- 5.2 Show how to construct a signal of hidden oscillations that grows in an unstable fashion. Where in the s -plane are the poles of the transforms of your signal(s)?
- 5.3 A first-order hold is a device that extrapolates a line over the interval from kT to $(k+1)T$ with slope given by the samples $r(kT - T)$ and $r(kT)$ starting from $r(kT)$ as shown in Fig. 5.19. Compute the frequency response of the first-order hold.
- 5.4 Consider the circuit of Fig. 5.20. By plotting the response to a signal that is zero for all sample instants except $t = 0$ and that is 1.0 at $t = 0$, show that this circuit implements a first-order hold.
- 5.5 Sketch the step response $y(t)$ of the system shown in Fig. 5.21 for $K = \frac{1}{2}$, 1, and 2.
- 5.6 Sketch the response of a *second-order* hold circuit to a step unit. What might be the major disadvantage of this data extrapolator?
- 5.7 A triangle hold is a device that has an output, as sketched in Fig. 5.22 that connects the samples of an input with straight lines.
- Sketch the impulse response of the triangle hold. Notice that it is noncausal.
 - Compute the transfer function of the hold.
 - Use MATLAB to plot the frequency response of the triangle hold.

Figure 5.19
Impulse response of a
first-order hold filter

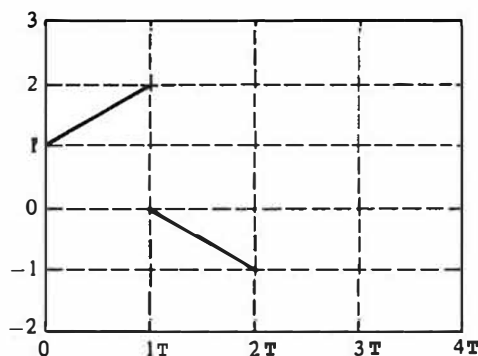


Figure 5.20
Block diagram of a
sample and first-order
hold

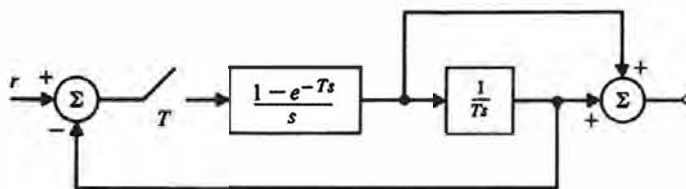


Figure 5.21
A sampled-data system

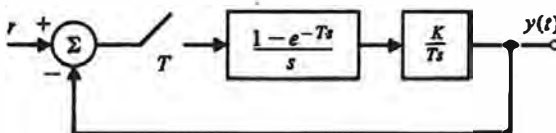
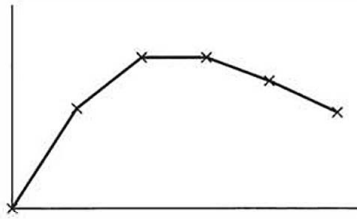
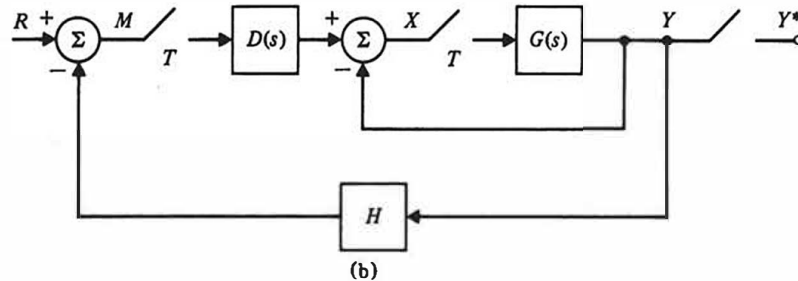
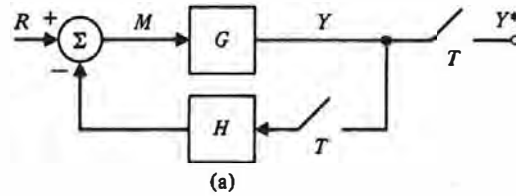


Figure 5.22
Response of a sample
and triangle hold



- (d) How would the frequency response be changed if the triangle hold is made to be causal by adding a delay of one sample period?
- 5.8 Sketch the output of a sample and zero-order hold to
- A step input.
 - A ramp input.
 - A sinusoid of frequency $\omega_s/10$.
- 5.9 Sketch the output of a sample and first-order hold to
- A step input.
 - A ramp input.
 - A sinusoid of frequency $\omega_s/10$.
- 5.10 Sketch the output of a sample and triangle hold to
- A step input.
 - A ramp input.
 - A sinusoid of frequency $\omega_s/10$.
- 5.11 Sketch the output of a sample and *causal* triangle hold to
- A step input.
 - A ramp input.
 - A sinusoid of frequency $\omega_s/10$.
- 5.12 A sinusoid of frequency 11 rad/sec. is sampled at the frequency $\omega_s = 5$ rad/sec.
- Indicate the component frequencies up to $\omega = 20$ rad/sec.
 - Indicate the relative amplitudes of the components up to 20 rad/sec. if the sampler is followed by a zero-hold.
- 5.13 A signal $r(t) = \sin(2t) + \sin(15t)$ is sampled at the frequency $\omega_s = 16$.
- Indicate the frequency of the components in the sampled signal up to $\omega = 32$.
 - Indicate the relative amplitudes of the signals in the output if the signal is passed through the anti-aliasing filter with transfer function $\frac{1}{(\frac{\omega}{\omega_s} + 1)^2}$ before sampling. You can use MATLAB to compute the filter gain.
- 5.14 Derive Eq. (5.38).
- 5.15 Find the transform of the output $Y(s)$ and its samples $Y^*(s)$ for the block diagrams shown in Fig. 5.23. Indicate whether a transfer function exists in each case.
- 5.16 Assume the following transfer functions are preceded by a sampler and zero-order hold and followed by a sampler. Compute the resulting discrete transfer functions.

Figure 5.23Block diagrams of
sampled data systems.(a) Single loop;
(b) multiple loop

- (a) $G_1(s) = 1/s^2$
 (b) $G_2(s) = e^{-1.5s}/(s+1)$
 (c) $G_3(s) = 1/s(s+1)$
 (d) $G_4(s) = e^{-1.5s}/s(s+1)$
 (e) $G_5(s) = 1/(s^2-1)$

5.17 One technique for examining the response of a sampled data system between sampling instants is to shift the response a fraction of a period to the left and to sample the result. The effect is as shown in the block diagram of Fig. 5.24 and described by the equation

$$Y^*(s, m) = R^*(s) \{G(s)e^{mTs}\}^*.$$

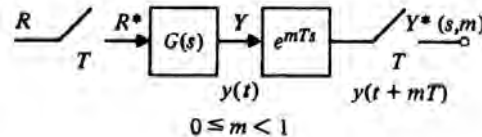
As a function of z , the equivalent equation is

$$Y(z) = R(z)G(z, m).$$

The function $G(z, m)$ is called the modified z -transform of $G(s)$. In the figure, let

$$G(s) = \frac{1}{(s+1)}, \quad R(s) = \frac{1}{s}, \quad \text{and} \quad T = 1.$$

- (a) Compute $y(t)$ by calculating $y(kT)$ from the ordinary z -transform and observing that between samples, the output $y(t)$ is an exponential decay with

Figure 5.24Block diagrams showing
the modified z -transform

unit time constant. Sketch the response for five sample intervals. Notice that this technique is the essence of the partial-fraction method of obtaining the ripple.

- (b) Compute the modified z -transform for $m = \frac{1}{2}$ and compute the samples according to the equation for $Y(z, m)$. Plot these on the same plot as that of $y(t)$ and verify that you have found the values at the mid-points of the sampling pattern.

5.8 Appendix

To compute the transform of a sinusoid, we consider first the Fourier transform of $v(t) = e^{j\omega_o t + j\phi}$. For this we have

$$V(j\omega) = \int_{-\infty}^{\infty} e^{(j\omega_o t + j\phi)} e^{-j\omega t} dt. \quad (5.59)$$

This integral does not converge in any obvious way, but we can approach it from the back door, as it were. Consider again the impulse, $\delta(t)$. The direct transform of this object is easy, considering the sifting property, as follows

$$\int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt = 1.$$

Now the general form of the inverse Fourier transform is given by

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(j\omega) e^{j\omega t} d\omega.$$

If we apply the inverse transform integral to the impulse and its transform, we take $f(t) = \delta(t)$ and $F(j\omega) = 1$ with the result

$$\delta(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{j\omega t} d\omega.$$

However, except for notation and a simple change of variables, this is exactly the integral we needed to evaluate the spectrum of the single exponential. If we exchange t with ω the integral reads

$$\delta(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{j\omega t} dt.$$

Eq. (5.59) is of this form

$$\begin{aligned} V(j\omega) &= \int_{-\infty}^{\infty} e^{(j\omega_o t + j\phi)} e^{-j\omega t} dt \\ &= e^{j\phi} \int_{-\infty}^{\infty} e^{jt(\omega_o - \omega)} dt \\ &= 2\pi e^{j\phi} \delta(\omega - \omega_o). \end{aligned}$$

At the last step in this development, the sign of the argument in the delta function was changed, which is legal because $\delta(t)$ is an even function and $\delta(t) = \delta(-t)$. The argument is more natural as $(\omega - \omega_o)$ rather than the opposite.

• 6 •

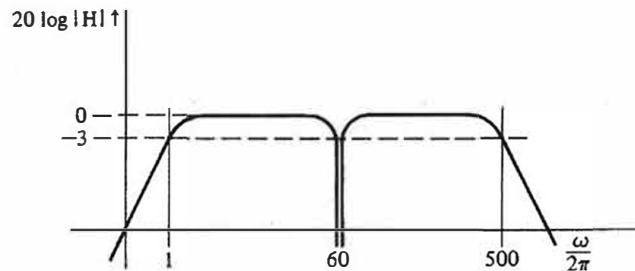
Discrete Equivalents

A Perspective on Computing Discrete Equivalents

One of the exciting fields of application of digital systems¹ is in signal processing and digital filtering. A filter is a device designed to pass desirable signal components and to reject undesirable ones; in signal processing it is common to represent signals as a sum of sinusoids and to define the “desirable components” as those signals whose frequencies are in a specified band. Thus a radio receiver is designed to pass the band of frequencies transmitted by the station we want to hear and reject all others. We would call a filter which does this a **bandpass filter**. In electrocardiography it often happens that power-line frequency signals are strong and unwanted, so a filter is designed to pass signals between 1 and 500 Hz but to eliminate those at 60 Hz. The magnitude of the transfer function for this purpose may look like Fig. 6.1 on a log-frequency scale, where the amplitude response between 59.5 and 60.5 Hz might reach 10^{-3} . Here we have a band-reject filter with a 60-dB rejection ratio in a 1-Hz band centered at 60 Hz.

In long-distance telephony some filters play a conceptually different role. There the issue is that ideal transmission requires that all desired frequencies be

Figure 6.1
Magnitude of a
low-frequency bandpass
filter with a narrow
rejection band



¹ Including microprocessors and special-purpose devices for digital signal processing, called DSP chips.

treated equally but transmission media—wires or microwaves—introduce distortion in the amplitude and phase of the sinusoids that comprise the desired signal and this distortion must be removed. Filters to correct the distortion are called **equalizers**. Finally, the dynamic response of control systems requires modification in order for the complete system to have satisfactory dynamic response. We call the devices that make these changes **compensators**.

Whatever the name—filter, equalizer, or compensator—many fields have use for linear dynamic systems having a transfer function with specified characteristics of amplitude and phase. Increasingly the power and flexibility of digital processors makes it attractive to perform these functions by digital means. The design of continuous electronic filters is a well-established subject that includes not only very sophisticated techniques but also well-tested computer programs to carry out the designs [Van Valkenburg (1982)]. Consequently, an important approach to digital filter design is to start with a good analog design and construct a filter having a discrete frequency response that approximates that of the satisfactory design. For digital control systems we have much the same motivation: Continuous-control designs are well established and one can take advantage of a good continuous design by finding a discrete equivalent to the continuous compensator. This method of design is called **emulation**. Although much of our presentation in this book is oriented toward direct digital design and away from emulation of continuous designs with digital equivalents, it is important to understand the techniques of discrete equivalents both for purposes of comparison and because it is widely used by practicing engineers.

emulation

Chapter Overview

The specific problem of this chapter is to find a discrete transfer function that will have approximately the same characteristics over the frequency range of importance as a given transfer function, $H(s)$. Three approaches to this task are presented. The first method is based on *numerical integration* of the differential equations that describe the given design. While there are many techniques for numerical integration, only simple formulas based on rectangular and trapezoid rules are presented. The second approach is based on comparisons of the s and z domains. Note that the natural response of a continuous filter with a pole at some point $s = s_0$ will, when sampled with period T , represent the response of a discrete filter with a pole at $z = e^{s_0 T}$. This formula can be used to map the poles and zeros of the given design into poles and zeros of an approximating discrete filter. This is called *pole and zero mapping*. The third and final approach is based on taking the samples of the input signal, extrapolating between samples to form an approximation to the signal, and passing this approximation through the given filter transfer function. This technique is called *hold equivalence*. The methods are compared with respect to the quality of the approximation in the frequency domain as well as the ease of computation of the designs.

6.1 Design of Discrete Equivalents via Numerical Integration

The topic of numerical integration of differential equations is quite complex, and only the most elementary techniques are presented here. For example, we only consider formulas of low complexity and fixed step-size. The fundamental concept is to represent the given filter transfer function $H(s)$ as a differential equation and to derive a difference equation whose solution is an approximation of the differential equation. For example, the system

$$\frac{U(s)}{E(s)} = H(s) = \frac{a}{s + a} \quad (6.1)$$

is equivalent to the differential equation

$$\dot{u} + au = ae. \quad (6.2)$$

Now, if we write Eq. (6.2) in integral form, we have a development much like that described in Chapter 4, except that the integral is more complex here

$$\begin{aligned} u(t) &= \int_0^t [-au(\tau) + ae(\tau)] d\tau, \\ u(kT) &= \int_0^{kT-T} [-au + ae] d\tau + \int_{kT-T}^{kT} [-au + ae] d\tau \\ &= u(kT - T) + \left\{ \begin{array}{l} \text{area of } -au + ae \\ \text{over } kT - T \leq \tau < kT. \end{array} \right. \end{aligned} \quad (6.3)$$

Many rules have been developed based on how the incremental area term is approximated. Three possibilities are sketched in Fig. 6.2. The first approximation leads to the **forward rectangular rule**² wherein we approximate the area by the rectangle looking forward from $kT - T$ and take the amplitude of the rectangle to be the value of the integrand at $kT - T$. The width of the rectangle is T . The result is an equation in the first approximation, u_1

$$\begin{aligned} u_1(kT) &= u_1(kT - T) + T[-au_1(kT - T) + ae(kT - T)] \\ &= (1 - aT)u_1(kT - T) + aTe(kT - T). \end{aligned} \quad (6.4)$$

The transfer function corresponding to the forward rectangular rule in this case is

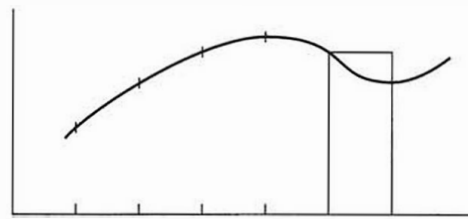
$$\begin{aligned} H_F(z) &= \frac{aTz^{-1}}{1 - (1 - aT)z^{-1}} \\ &= \frac{a}{(z - 1)/T + a} \quad (\text{forward rectangular rule}). \end{aligned} \quad (6.5)$$

² Also known as *Euler's rule*.

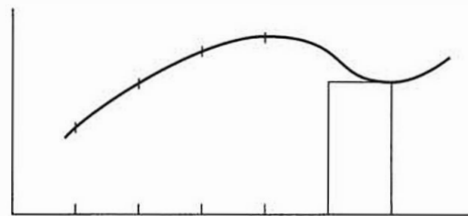
Figure 6.2

Sketches of three ways the area under the curve from kT to $kT + T$ can be approximated:

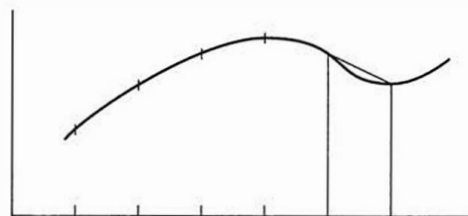
(a) forward rectangular rule, (b) backward rectangular rule, (c) trapezoid rule



(a) Forward Rectangular Rule



(b) Backward Rectangular Rule



(c) Trapezoid Rule

A second rule follows from taking the amplitude of the approximating rectangle to be the value looking *backward* from kT toward $kT - T$, namely, $-au(kT) + ae(kT)$. The equation for u_2 , the second approximation,³ is

$$\begin{aligned} u_2(kT) &= u_2(kT - T) + T[-au_2(kT) + ae(kT)] \\ &= \frac{u_2(kT - T)}{1 + aT} + \frac{aT}{1 + aT} e(kT). \end{aligned} \quad (6.6)$$

³ It is worth noting that in order to solve for Eq. (6.6) we had to eliminate $u(kT)$ from the right-hand side where it entered from the integrand. Had Eq. (6.2) been nonlinear, the result would have been an implicit equation requiring an iterative solution. This topic is the subject of predictor-corrector rules, which are beyond our scope of interest. A discussion is found in most books on numerical analysis. See Golub and Van Loan (1983).

Again we take the z -transform and compute the transfer function of the **backward rectangular rule**

$$\begin{aligned} H_b(z) &= \frac{aT}{1+aT} \frac{1}{1-z^{-1}/(1+aT)} = \frac{aTz}{z(1+aT)-1} \\ &= \frac{a}{(z-1)/Tz+a} \quad (\text{backward rectangular rule}). \end{aligned} \quad (6.7)$$

Our final version of integration rules is the **trapezoid rule** found by taking the area approximated in Eq. (6.3) to be that of the trapezoid formed by the average of the previously selected rectangles. The approximating difference equation is

$$\begin{aligned} u_3(kT) &= u_3(kT-T) + \frac{T}{2} [-au_3(kT-T) \\ &\quad + ae(kT-T) - au_3(kT) + ae(kT)] \\ &= \frac{1-(aT/2)}{1+(aT/2)} u_3(kT-T) + \frac{aT/2}{1+(aT/2)} [e_3(kT-T) + e_3(kT)]. \end{aligned} \quad (6.8)$$

The corresponding transfer function from the trapezoid rule is

$$\begin{aligned} H_T(z) &= \frac{aT(z+1)}{(2+aT)z+aT-2} \\ &= \frac{a}{(2/T)[(z-1)/(z+1)]+a} \quad (\text{trapezoid rule}). \end{aligned} \quad (6.9)$$

Suppose we tabulate our results obtained thus far.

$H(s)$	Method	Transfer function
$\frac{a}{s+a}$	Forward rule	$H_F = \frac{a}{(z-1)/T+a}$
$\frac{a}{s+a}$	Backward rule	$H_B = \frac{a}{(z-1)/Tz+a}$
$\frac{a}{s+a}$	Trapezoid rule	$H = \frac{a}{(2/T)\{(z-1)/(z+1)\}+a}$

(6.10)

From direct comparison of $H(s)$ with the three approximations in this tabulation, we can see that the effect of each of our methods is to present a discrete transfer function that can be obtained from the given Laplace transfer function

$H(s)$ by substitution of an approximation for the frequency variable as shown below

Method	Approximation
Forward rule	$s \leftarrow \frac{z-1}{T}$
Backward Rule	$s \leftarrow \frac{z-1}{Tz}$
Trapezoid Rule	$s \leftarrow \frac{2z-1}{Tz+1}$

(6.11)

The trapezoid-rule substitution is also known, especially in digital and sampled-data control circles, as **Tustin's method** [Tustin (1947)] after the British engineer whose work on nonlinear circuits stimulated a great deal of interest in this approach. The transformation is also called the **bilinear transformation** from consideration of its mathematical form. The design method can be summarized by stating the rule: Given a continuous transfer function (filter), $H(s)$, a discrete equivalent can be found by the substitution

$$H_T(z) = H(s)|_{s=\frac{2}{T}\frac{z-1}{z+1}}. \quad (6.12)$$

Each of the approximations given in Eq. (6.11) can be viewed as a map from the s -plane to the z -plane. A further understanding of the maps can be obtained by considering them graphically. For example, because the $(s = j\omega)$ -axis is the boundary between poles of stable systems and poles of unstable systems, it would be interesting to know how the $j\omega$ -axis is mapped by the three rules and where the left (stable) half of the s -plane appears in the z -plane. For this purpose we must solve the relations in Eq. (6.11) for z in terms of s . We find

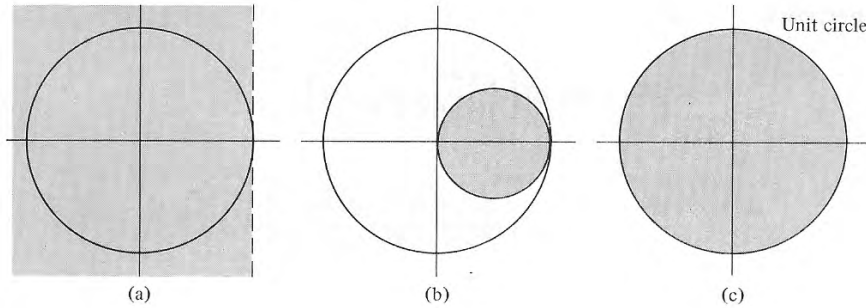
$$\begin{aligned} \text{i) } z &= 1 + Ts, & (\text{forward rectangular rule}). \\ \text{ii) } z &= \frac{1}{1 - Ts}, & (\text{backward rectangular rule}). \\ \text{iii) } z &= \frac{1 + Ts/2}{1 - Ts/2} & (\text{bilinear rule}). \end{aligned} \quad (6.13)$$

If we let $s = j\omega$ in these equations, we obtain the boundaries of the regions in the z -plane which originate from the stable portion of the s -plane. The shaded areas sketched in the z -plane in Fig. 6.3 are these stable regions for each case. To show that rule (ii) results in a circle, $\frac{1}{2}$ is added to and subtracted from the right-hand side to yield

$$\begin{aligned} z &= \frac{1}{2} + \left\{ \frac{1}{1 - Ts} - \frac{1}{2} \right\} \\ &= \frac{1}{2} - \frac{1}{2} \frac{1 + Ts}{1 - Ts}. \end{aligned} \quad (6.14)$$

Figure 6.3

Maps of the left-half of the s -plane by the integration rules of Eq. (6.10) into the z -plane. Stable s -plane poles map into the shaded regions in the z -plane. The unit circle is shown for reference. (a) Forward rectangular rule. (b) Backward rectangular rule. (c) Trapezoid or bilinear rule



Now it is easy to see that with $s = j\omega$, the magnitude of $z - \frac{1}{2}$ is constant

$$\left|z - \frac{1}{2}\right| = \frac{1}{2},$$

and the curve is thus a circle as drawn in Fig. 6.3(b). Because the unit circle is the stability boundary in the z -plane, it is apparent from Fig. 6.3 that the forward rectangular rule could cause a stable continuous filter to be mapped into an unstable digital filter.

It is especially interesting to notice that the bilinear rule maps the stable region of the s -plane exactly into the stable region of the z -plane although the entire $j\omega$ -axis of the s -plane is compressed into the 2π -length of the unit circle! Obviously a great deal of distortion takes place in the mapping in spite of the congruence of the stability regions. As our final rule deriving from numerical integration ideas, we discuss a formula that extends Tustin's rule one step in an attempt to correct for the inevitable distortion of real frequencies mapped by the rule. We begin with our elementary transfer function Eq. (6.1) and consider the bilinear rule approximation

$$H_T(z) = \frac{a}{(2/T)[(z-1)/(z+1)] + a}.$$

The original $H(s)$ had a pole at $s = -a$, and for real frequencies, $s = j\omega$, the magnitude of $H(j\omega)$ is given by

$$\begin{aligned} |H(j\omega)|^2 &= \frac{a^2}{\omega^2 + a^2} \\ &= \frac{1}{\omega^2/a^2 + 1}. \end{aligned}$$

Thus our reference filter has a half-power point, $|H|^2 = \frac{1}{2}$, at $\omega = a$. It will be interesting to know where $H_T(z)$ has a half-power point.

As we saw in Chapter 4, signals with poles on the imaginary axis in the s -plane (sinusoids) map into signals on the unit circle of the z -plane. A sinusoid of frequency ω_1 corresponds to $z_1 = e^{j\omega_1 T}$, and the response of $H_T(z)$ to a sinusoid of frequency ω_1 is $H_T(z_1)$. We consider now Eq. (6.8) for $H_T(z_1)$ and manipulate it into a more convenient form for our present purposes

$$\begin{aligned} H_T(z_1) &= a / \left(\frac{2}{T} \frac{e^{j\omega_1 T} - 1}{e^{j\omega_1 T} + 1} + a \right) \\ &= a / \left(\frac{2}{T} \frac{e^{j\omega_1 T/2} - e^{-j\omega_1 T/2}}{e^{j\omega_1 T/2} + e^{-j\omega_1 T/2}} + a \right) \\ &= a / \left(\frac{2}{T} j \tan \frac{\omega_1 T}{2} + a \right). \end{aligned} \quad (6.15)$$

The magnitude squared of H_T will be $\frac{1}{2}$ when

$$\frac{2}{T} \tan \frac{\omega_1 T}{2} = a$$

or

$$\tan \frac{\omega_1 T}{2} = \frac{aT}{2}. \quad (6.16)$$

Equation (6.16) is a measure of the frequency distortion or warping caused by Tustin's rule. Whereas we wanted to have a half-power point at $\omega = a$, we realized a half-power point at $\omega_1 = (2/T) \tan^{-1}(aT/2)$. ω_1 will be approximately correct only if $aT/2 \ll 1$ so that $\tan^{-1}(aT/2) \cong aT/2$, that is, if $\omega_s (= 2\pi/T) \gg a$ and the sample rate is much faster than the half-power frequency. We can turn our intentions around and suppose that we really want the half-power point to be at ω_1 . Equation (6.16) can be made into an equation of prewarping: If we select a according to Eq. (6.16), then, using Tustin's bilinear rule for the design, the half-power point will be at ω_1 . A statement of a complete set of rules for filter design via bilinear transformation with prewarping is

1. Write the desired filter characteristic with transform variable s and critical frequency ω_1 in the form $H(s/\omega_1)$.⁴
2. Replace ω_1 by a such that

$$a = \frac{2}{T} \tan \frac{\omega_1 T}{2},$$

⁴ The critical frequency need not be the band edge. We can use the band center of a bandpass filter or the crossover frequency of a Bode plot compensator. However, we must have $\omega_1 < \pi/T$ if a stable filter is to remain stable after warping.

and in place of $H(s/\omega_1)$, consider the prewarped function $H(s/a)$. For more complicated shapes, such as bandpass filters, the specification frequencies, such as band edges and center frequency, should be prewarped before the continuous design is done; and then the bilinear transformation will bring all these points to their correct frequencies in the digital filter.

3. Substitute

$$s = \frac{2}{T} \frac{z-1}{z+1}$$

in $H(s/a)$ to obtain the prewarped equivalent $H_p(z)$.

As a frequency substitution the result can be expressed as

$$H_p(z) = H\left(\frac{s}{\omega_1}\right) \bigg|_{\frac{s}{\omega_1} = \frac{1}{\tan(\omega_1 T/2)} \frac{z-1}{z+1}} \quad (6.17)$$

It is clear from Eq. (6.17) that when $\omega = \omega_1$, $H_p(z_1) = H(j1)$ and the discrete filter has exactly the same transmission at ω_1 as the continuous filter has at this frequency. This is the consequence of prewarping. We also note that as the sampling period gets small, $H_p(z)$ approaches $H(j\omega/\omega_1)$.

◆ Example 6.1 Computing a Discrete Equivalent

The transfer function of a third order low-pass Butterworth filter⁵ designed to have unity pass bandwidth ($\omega_p = 1$) is

$$H(s) = \frac{1}{s^3 + 2s^2 + 2s + 1}.$$

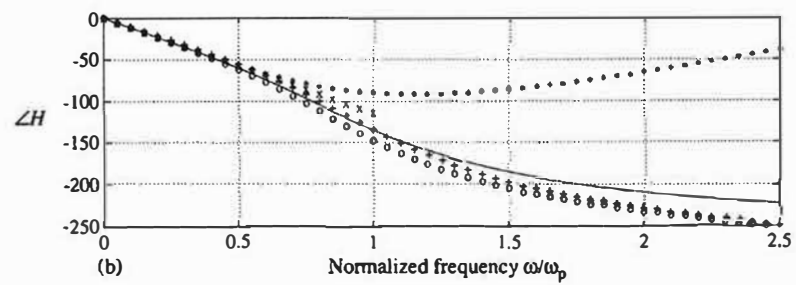
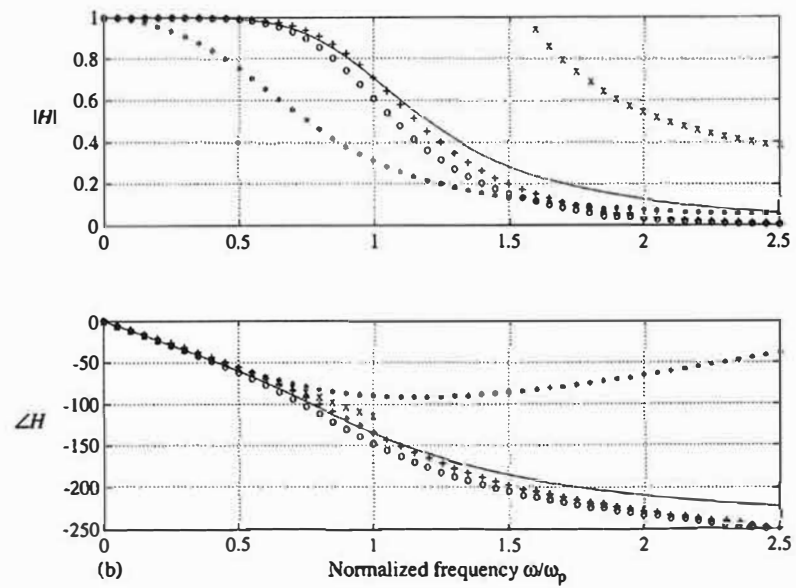
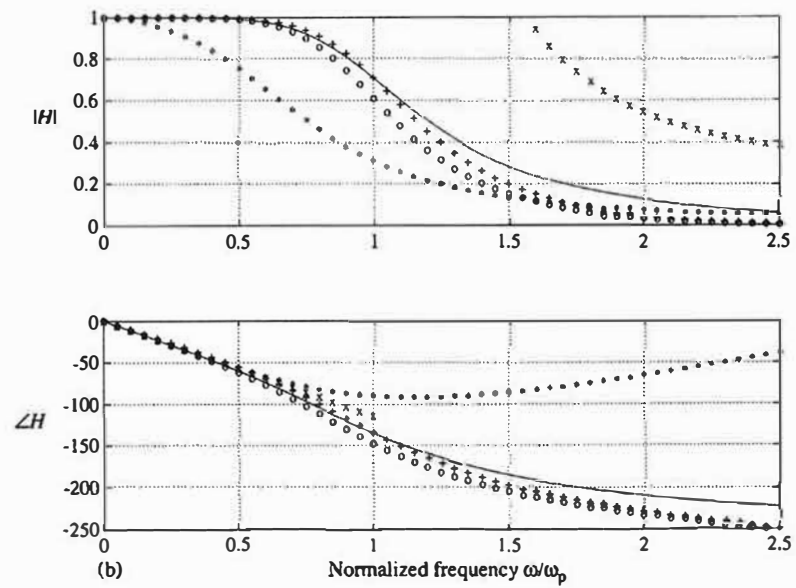
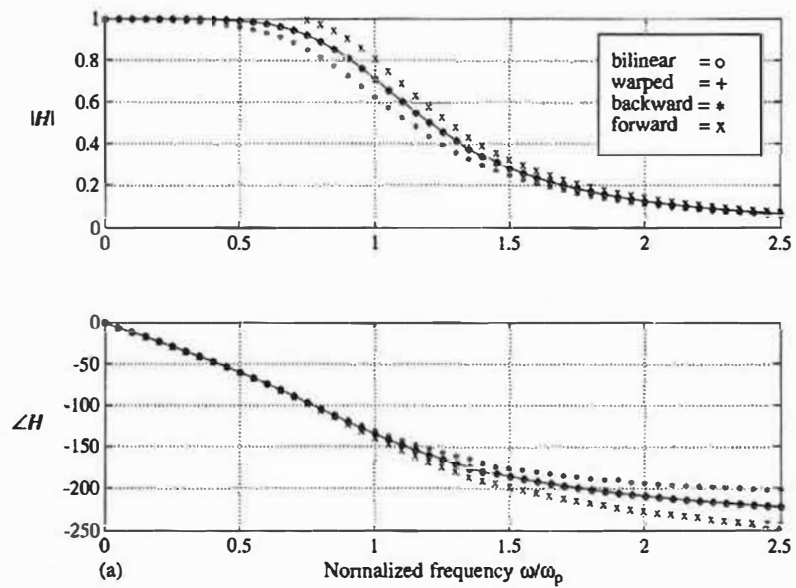
A simple frequency scaling would of course translate the design to have any desired passband frequency. Compute the discrete equivalents and plot the frequency responses using the forward rectangular rule, the backward rectangular rule, the Tustin bilinear rule and the bilinear rule with prewarping at $\omega = 1$. Use sampling periods $T = 0.1$, $T = 1$, and $T = 2$.

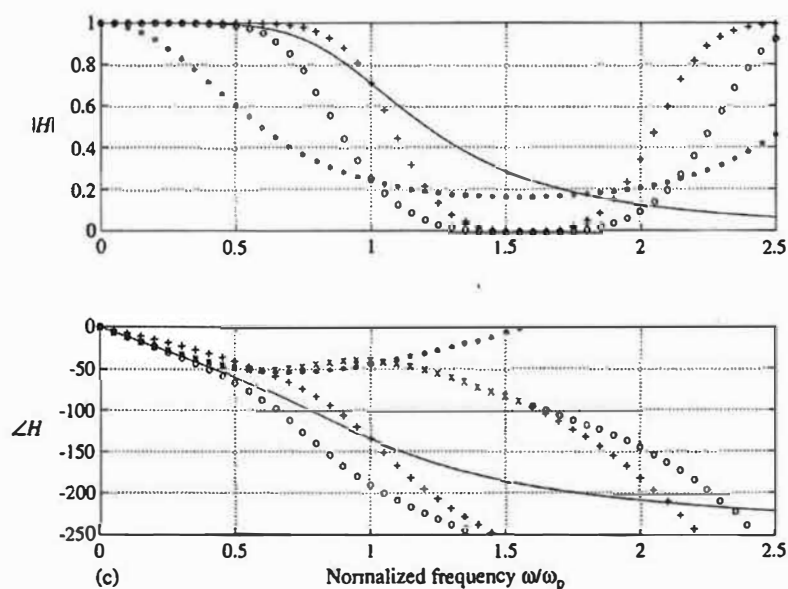
Solution. Computation of the discrete equivalents is numerically tedious and the state-space algorithms described below were used in MATLAB to generate the transfer functions and the response curves plotted in Fig. 6.4. Fig 6.4(a) shows that at a high sample rate ($T = 0.1$), where the ratio of sampling frequency to passband frequency is $\omega_s/\omega_p \approx 63$, all the rules do reasonably well but the rectangular rules are already showing some deviation. From Fig. 6.4(b) we see that at $\omega_s/\omega_p = 2\pi$ the rectangular rules are useless (the forward rule is unstable). Finally, in Fig. 6.4(c) at very slow sampling frequency with $\omega_s/\omega_p = \pi$ corresponding to a sampling period of $T = 2$ sec, only with prewarping do we have a design that comes even close to the continuous response. In each case at the Nyquist frequency, $\omega = \pi/T$, the magnitude

⁵ A description of the properties of Butterworth filters is given in most books on filter design and briefly in Franklin, Powell and Emami-Naeini (2019).

Figure 6.4

(a) Response of third-order lowpass filter and digital equivalents for $\omega_s/\omega_p = 20\pi$.
 (b) Response of third-order lowpass filter and digital equivalents for $\omega_s/\omega_p = 2\pi$.
 (c) Response of third-order lowpass filter and digital equivalents for $\omega_s/\omega_p = \pi$





responses of the discrete filters start to repeat according to the periodic nature of discrete-transfer-function frequency responses. It can be seen that the magnitude and phase of the prewarped designs match those of the continuous filter exactly at the band edge, $\omega = 1$, for all these cases. This is no surprise, because such matching was the whole idea of prewarping.

The formulas for discrete equivalents are particularly simple and convenient when expressed in state-variable form and used with a computer-aided design package. For example, suppose we have a vector-matrix description of a continuous design in the form of the equations

$$\begin{aligned}\dot{\mathbf{x}} &= \mathbf{A}\mathbf{x} + \mathbf{B}e, \\ u &= \mathbf{C}\mathbf{x} + \mathbf{D}e.\end{aligned}\tag{6.18}$$

The Laplace transform of this equation is

$$\begin{aligned}s\mathbf{X} &= \mathbf{A}\mathbf{X} + \mathbf{B}E, \\ U &= \mathbf{C}\mathbf{X} + \mathbf{D}E.\end{aligned}\tag{6.19}$$

We can now substitute for s in Eq. (6.19) any of the forms in z corresponding to an integration rule. For example, the forward rectangular rule is to replace s with $(z - 1)/T$ from Eq. (6.11)

$$\begin{aligned}\frac{z - 1}{T}\mathbf{X} &= \mathbf{A}\mathbf{X} + \mathbf{B}E, \\ U &= \mathbf{C}\mathbf{X} + \mathbf{D}E.\end{aligned}\tag{6.20}$$

In the time domain, the operator z corresponds to forward shift; that is, $zx(k) = x(k+1)$. Thus the corresponding discrete equations in the time domain are

$$\begin{aligned} \mathbf{x}(k+1) - \mathbf{x}(k) &= T\mathbf{A}\mathbf{x}(k) + T\mathbf{B}e(k), \\ \mathbf{x}(k+1) &= (\mathbf{I} + T\mathbf{A})\mathbf{x}(k) + T\mathbf{B}e(k), \\ u &= \mathbf{C}\mathbf{x} + \mathbf{D}e. \end{aligned} \quad (6.21)$$

Equation (6.21) is a state-space formula for the forward rule equivalent.

For the backward rule, substitute $s \leftarrow (z-1)/zT$ with the result

$$\frac{z-1}{Tz}\mathbf{X} = \mathbf{A}\mathbf{X} + \mathbf{B}E,$$

which corresponds to the time domain equations

$$\mathbf{x}(k+1) - \mathbf{x}(k) = T\mathbf{A}\mathbf{x}(k+1) + T\mathbf{B}e(k+1). \quad (6.22)$$

In this equation, there are terms in $k+1$ on both the right- and left-hand sides. In order to get an equation with such terms only on the left, transpose all $k+1$ terms to the left and define them as a new state vector

$$\begin{aligned} \mathbf{x}(k+1) - T\mathbf{A}\mathbf{x}(k+1) - T\mathbf{B}e(k+1) &= \mathbf{x}(k) \\ &\triangleq \mathbf{w}(k+1). \end{aligned} \quad (6.23)$$

From this equation, solving for $\mathbf{x}$ in terms of $\mathbf{w}$ and e

$$\begin{aligned} (\mathbf{I} - T\mathbf{A})\mathbf{x} &= \mathbf{w} + T\mathbf{B}e \\ \mathbf{x} &= (\mathbf{I} - T\mathbf{A})^{-1}\mathbf{w} + (\mathbf{I} - T\mathbf{A})^{-1}T\mathbf{B}e. \end{aligned} \quad (6.24)$$

With this expression for $\mathbf{x}$, Eq. (6.23) can be put in standard form as

$$\mathbf{w}(k+1) = (\mathbf{I} - T\mathbf{A})^{-1}\mathbf{w}(k) + (\mathbf{I} - T\mathbf{A})^{-1}T\mathbf{B}e(k), \quad (6.25)$$

and the output equation is now

$$u(k) = \mathbf{C}(\mathbf{I} - T\mathbf{A})^{-1}\mathbf{w} + \{\mathbf{D} + \mathbf{C}(\mathbf{I} - T\mathbf{A})^{-1}T\mathbf{B}\}e. \quad (6.26)$$

Equation (6.25) plus Eq. (6.26) are a state-space description of the backward rule equivalent to Eq. (6.18).

Finally, for the trapezoid or bilinear rule, the z -transform equivalent is obtained from

$$\begin{aligned} \frac{2(z-1)}{T(z+1)}\mathbf{X} &= \mathbf{A}\mathbf{X} + \mathbf{B}E \\ (z-1)\mathbf{X} &= \frac{\mathbf{A}T}{2}(z+1)\mathbf{X} + \frac{\mathbf{B}T}{2}(z+1)E \\ U &= \mathbf{C}\mathbf{X} + \mathbf{D}E, \end{aligned} \quad (6.27)$$

and the time domain equation for the state is

$$\mathbf{x}(k+1) - \mathbf{x}(k) = \frac{\mathbf{A}T}{2}(\mathbf{x}(k+1) + \mathbf{x}(k)) + \frac{\mathbf{B}T}{2}(e(k+1) + e(k)). \quad (6.28)$$

Once more, collect all the $k + 1$ terms onto the left and define these as $\mathbf{w}(k + 1)$ as follows⁶

$$\begin{aligned} \mathbf{x}(k + 1) - \frac{\mathbf{A}T}{2}\mathbf{x}(k + 1) - \frac{\mathbf{B}T}{2}e(k + 1) &= \mathbf{x}(k) + \frac{\mathbf{A}T}{2}\mathbf{x}(k) + \frac{\mathbf{B}T}{2}e(k) \\ &\triangleq \sqrt{T}\mathbf{w}(k + 1). \end{aligned} \quad (6.29)$$

Writing the definition of $\mathbf{w}$ at time k , solve for $\mathbf{x}$ as before

$$\begin{aligned} \left(\mathbf{I} - \frac{\mathbf{A}T}{2}\right)\mathbf{x} &= \sqrt{T}\mathbf{w} + \frac{\mathbf{B}T}{2}e \\ \mathbf{x} &= \left(\mathbf{I} - \frac{\mathbf{A}T}{2}\right)^{-1} \sqrt{T}\mathbf{w} + \left(\mathbf{I} - \frac{\mathbf{A}T}{2}\right)^{-1} \frac{\mathbf{B}T}{2}e. \end{aligned} \quad (6.30)$$

Substituting Eq. (6.30) into Eq. (6.29), we obtain

$$\begin{aligned} \sqrt{T}\mathbf{w}(k + 1) &= \left(\mathbf{I} + \frac{\mathbf{A}T}{2}\right) \left(\mathbf{I} - \frac{\mathbf{A}T}{2}\right)^{-1} \left\{ \sqrt{T}\mathbf{w}(k) + \frac{\mathbf{B}T}{2}e \right\} + \frac{\mathbf{B}T}{2}e(k) \\ \mathbf{w}(k + 1) &= \left(\mathbf{I} + \frac{\mathbf{A}T}{2}\right) \left(\mathbf{I} - \frac{\mathbf{A}T}{2}\right)^{-1} \mathbf{w}(k) \\ &\quad + \left\{ \left(\mathbf{I} + \frac{\mathbf{A}T}{2}\right) \left(\mathbf{I} - \frac{\mathbf{A}T}{2}\right)^{-1} + \mathbf{I} \right\} \frac{\mathbf{B}\sqrt{T}}{2}e(k) \\ &= \left(\mathbf{I} + \frac{\mathbf{A}T}{2}\right) \left(\mathbf{I} - \frac{\mathbf{A}T}{2}\right)^{-1} \mathbf{w}(k) + \left(\mathbf{I} - \frac{\mathbf{A}T}{2}\right)^{-1} \mathbf{B}\sqrt{T}e(k). \end{aligned} \quad (6.31)$$

In following this algebra, it is useful to know that in deriving the last part of Eq. (6.31), we expressed the identity $\mathbf{I}$ as $(\mathbf{I} - \frac{\mathbf{A}T}{2})(\mathbf{I} - \frac{\mathbf{A}T}{2})^{-1}$ and factored out $(\mathbf{I} - \frac{\mathbf{A}T}{2})^{-1}$ on the right.

To obtain the output equation for the bilinear equivalent, we substitute Eq. (6.30) into the second part of Eq. (6.27):

$$u(k) = \sqrt{T}\mathbf{C} \left(\mathbf{I} - \frac{\mathbf{A}T}{2}\right)^{-1} \mathbf{w}(k) + \left\{ \mathbf{D} + \mathbf{C} \left(\mathbf{I} - \frac{\mathbf{A}T}{2}\right)^{-1} \frac{\mathbf{B}T}{2} \right\} e(k).$$

These results can be tabulated for convenient reference. Suppose we have a continuous system described by

$$\begin{aligned} \dot{\mathbf{x}}(t) &= \mathbf{A}\mathbf{x}(t) + \mathbf{B}e(t), \\ u(t) &= \mathbf{C}\mathbf{x}(t) + \mathbf{D}e(t). \end{aligned}$$

⁶ The scale factor of $\sqrt{T}$ is introduced so that the gain of the discrete equivalent will be balanced between input and output, a rather technical condition. See Al Saggar and Franklin (1986) for many more details.

Then a discrete equivalent at sampling period T will be described by the equations

$$\mathbf{w}(k+1) = \Phi \mathbf{w}(k) + \Gamma e(k),$$

$$u(k) = \mathbf{H} \mathbf{w}(k) + \mathbf{J} e(k),$$

where Φ , Γ , $\mathbf{H}$, and $\mathbf{J}$ are given respectively as follows:

	Forward	Backward	Bilinear
Φ	$\mathbf{I} + \mathbf{A}T$	$(\mathbf{I} - \mathbf{A}T)^{-1} \mathbf{B}T$	$(\mathbf{I} + \frac{\mathbf{A}T}{2})(\mathbf{I} - \frac{\mathbf{A}T}{2})^{-1}$
Γ	$\mathbf{B}T$	$(\mathbf{I} - \mathbf{A}T)^{-1}$	$(\mathbf{I} - \frac{\mathbf{A}T}{2})^{-1} \mathbf{B} \sqrt{T}$
$\mathbf{H}$		$\mathbf{C}(\mathbf{I} - \mathbf{A}T)^{-1}$	$\sqrt{T} \mathbf{C}(\mathbf{I} - \frac{\mathbf{A}T}{2})^{-1}$
$\mathbf{J}$	$\mathbf{D}$	$\mathbf{D} + \mathbf{C}(\mathbf{I} - \mathbf{A}T)^{-1} \mathbf{B}T$	$\mathbf{D} + \mathbf{C}(\mathbf{I} - \frac{\mathbf{A}T}{2})^{-1} \mathbf{B}T/2$

The MATLAB Control Toolbox provides for the computation of Tustin bilinear equivalents with the function `c2d`. The syntax of computing the bilinear equivalent SYSD of a continuous system SYS at sampling period T_s is

$$\text{SYSD} = \text{c2d}(\text{SYS}, T_s, \text{'tustin'}) \quad (6.32)$$

If 'tustin' is replaced with 'prewarp', the bilinear equivalent with prewarping is computed.

6.2 Zero-Pole Matching Equivalents

A very simple but effective method of obtaining a discrete equivalent to a continuous transfer function is to be found by extrapolation of the relation derived in Chapter 4 between the s - and z -planes. If we take the z -transform of samples of a continuous signal $e(t)$, then the poles of the discrete transform $E(z)$ are related to the poles of $E(s)$ according to $z = e^{sT}$. We must go through the z -transform process to locate the zeros of $E(z)$, however. The idea of the zero-pole matching technique is that the map $z = e^{sT}$ could reasonably be applied to the zeros also. The technique consists of a set of heuristic rules for locating the zeros and poles and setting the gain of a z -transform that will describe a discrete, equivalent transfer function that approximates the given $H(s)$. The rules are as follows:

1. All poles of $H(s)$ are mapped according to $z = e^{sT}$. If $H(s)$ has a pole at $s = -a$, then $H_{zp}(z)$ has a pole at $z = e^{-aT}$. If $H(s)$ has a pole at $-a + jb$, then $H_{zp}(z)$ has a pole at $re^{j\theta}$, where $r = e^{-aT}$ and $\theta = bT$.
2. All finite zeros are also mapped by $z = e^{sT}$. If $H(s)$ has a zero at $s = -a$, then $H_{zp}(z)$ has a zero at $z = e^{-aT}$, and so on.
3. The zeros of $H(s)$ at $s = \infty$ are mapped in $H_{zp}(z)$ to the point $z = -1$. The rationale behind this rule is that the map of real frequencies from $j\omega = 0$ to

increasing ω is onto the unit circle at $z = e^{j0} = 1$ until $z = e^{j\pi} = -1$. Thus the point $z = -1$ represents, in a real way, the highest frequency possible in the discrete transfer function, so it is appropriate that if $H(s)$ is zero at the highest (continuous) frequency, $|H_{zp}(z)|$ should be zero at $z = -1$, the highest frequency that can be processed by the digital filter.

- (a) If no delay in the discrete response is desired, all zeros at $s = \infty$ are mapped to $z = -1$.
 - (b) If one sample period delay is desired to give the computer time to complete the output calculation, then one of the zeros at $s = \infty$ is mapped to $z = \infty$ and the others mapped to $z = -1$. With this choice, $H_{zp}(z)$ is left with a number of finite zeros one fewer than the number of finite poles.
4. The gain of the digital filter is selected to match the gain of $H(s)$ at the band center or a similar critical point. In most control applications, the critical frequency is $s = 0$, and hence we typically select the gain so that

$$H(s)|_{s=0} = H_{zp}(z)|_{z=1}$$

A more complete discussion is contained in Section 8.3.3 in Franklin, Powell, Emami-Naeini, 7th or 8th edition.

◆ Example 6.2 A Zero-pole Matching Equivalent

Compute the discrete equivalent to

$$H(s) = \frac{a}{s + a}$$

by zero-pole matching.

Solution. The pole of $H(s)$ at $s = -a$ will map to a pole of $H(z)$ at e^{-aT} . The zero at $s = \infty$ will map to a zero at $z = -1$. The gain of $H(s)$ at $s = 0$ is 1. To match this gain in $H(z)$ at $z = 1$ requires gain of $\frac{1 - e^{-aT}}{2}$. The final function is given by

$$H_{zp}(z) = \frac{(z + 1)(1 - e^{-aT})}{2(z - e^{-aT})}, \quad (6.33)$$

or, using rule 3(b), the result is

$$H_{zp}(z) = \frac{1 - e^{-aT}}{z - e^{-aT}}. \quad (6.34)$$

As with the rules based on numerical analysis, an algorithm to generate the matched zero-pole equivalent is also readily constructed. In MATLAB, a matched

zero-pole equivalent, SYSD, at sample period T_s to the continuous system, SYS, is given by

$$\text{SYSD} = \text{c2d}(\text{SYS}, T_s, \text{'matched'}).$$

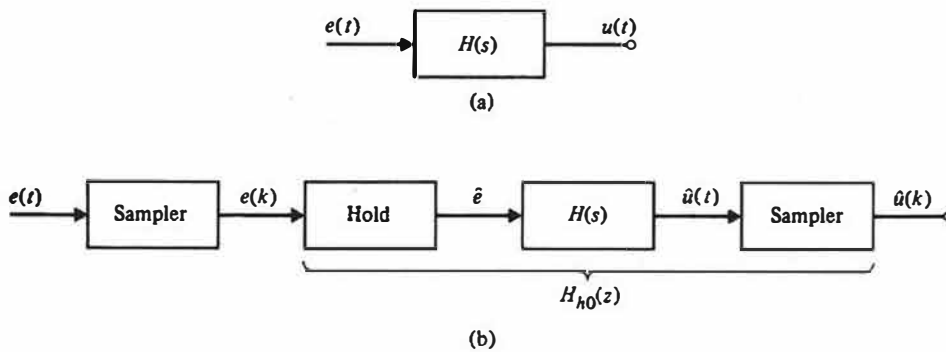
The frequency response of the matched zero-pole equivalent for the third-order Butterworth filter of Example 6.1 is plotted in Fig. 6.9 along with that of other equivalents for purposes of comparison.

6.3 Hold Equivalents

For this technique, we construct the situation sketched in Fig. 6.5. The samplers in Fig. 6.5(b) provide the samples at the input of $H_{ho}(z)$ and take samples at its output insuring that $H_{ho}(z)$ can be realized as a discrete transfer function. The philosophy of the design is the following. We are asked to design a discrete system that, with an input consisting of *samples* of $e(t)$, has an output that approximates the output of the continuous filter $H(s)$ whose input is the *continuous* $e(t)$. The discrete hold equivalent is constructed by first approximating $e(t)$ from the samples $e(k)$ with a hold filter and then putting this $\hat{e}(t)$ through the given $H(s)$. There are many techniques for taking a sequence of samples and extrapolating or holding them to produce a continuous signal.⁷ Suppose we have the $e(t)$ as sketched in Fig. 6.6. This figure also shows a sketch of a piecewise constant approximation to $e(t)$

Figure 6.5

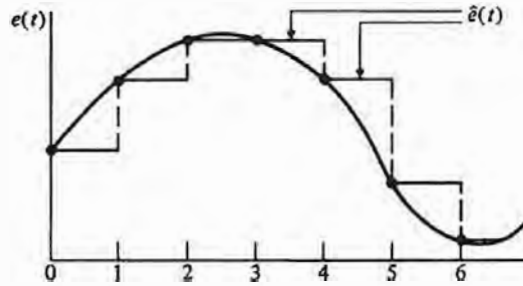
System construction for hold equivalents. (a) A continuous transfer function. (b) Block diagram of an equivalent system.



⁷ Some books on digital-signal processing suggest using no hold at all, using the equivalent $H(z) = \mathcal{Z}\{H(s)\}$. This choice is called the z -transform equivalent.

Figure 6.6

A signal, its samples, and its approximation by a zero-order hold



obtained by the operation of holding $e_h(t)$ constant at $e(k)$ over the interval from kT to $(k+1)T$. This operation is the *zero-order hold* (or ZOH) we've discussed before. If we use a first-order polynomial for extrapolation, we have a *first-order hold* (or FOH), and so on for higher-order holds.

6.3.1 Zero-Order Hold Equivalent

If the approximating hold is the zero-order hold, then we have for our approximation exactly the same situation that in Chapter 4 was analyzed as a sampled-data system.⁸ Therefore, the zero-order-hold equivalent to $H(s)$ is given by Eq. (4.41), which is

$$H_{h0}(z) = (1 - z^{-1}) \mathcal{Z} \left\{ \frac{H(s)}{s} \right\}. \quad (6.35)$$

◆ Example 6.3 A Hold Equivalent

Find the zero-order-hold equivalent to the first-order transfer function

$$H(s) = \frac{a}{s+a}.$$

Solution. The partial fraction expansion of the s -plane terms of Eq. (6.35) is

$$\frac{H(s)}{s} = \frac{a}{s(s+a)} = \frac{1}{s} - \frac{1}{s+a}$$

and the z -transform is

$$\mathcal{Z} \left\{ \frac{H(s)}{s} \right\} = \mathcal{Z} \left\{ \frac{1}{s} \right\} - \mathcal{Z} \left\{ \frac{1}{s+a} \right\}, \quad (6.36)$$

⁸ Recall that we noticed in Chapter 5 that the signal $\hat{e}$ is, on the average, delayed from e by $T/2$ sec. The size of this delay is one measure of the quality of the approximation and can be used as a guide to the selection of T .

and, by definition of the operation given in Eq. (6.36)

$$\begin{aligned} \mathcal{Z} \left\{ \frac{H(s)}{s} \right\} &= \sum_0^{\infty} z^{-k} - \sum_0^{\infty} z^{-k} e^{-akT} \\ &= \frac{1}{1-z^{-1}} - \frac{1}{1-e^{-aT}z^{-1}} \\ &= \frac{(1-e^{-aT}z^{-1}) - (1-z^{-1})}{(1-z^{-1})(1-e^{-aT}z^{-1})}. \end{aligned} \quad (6.37)$$

Substituting Eq. (6.37) in Eq. (6.35), the zero-order-hold equivalent of $H(s)$ is found as

$$H_{h0}(z) = \frac{(1-e^{-aT})}{z-e^{-aT}}. \quad (6.38)$$

We note that for the trivial example given, the zero-order-hold equivalent of Eq. (6.38) is identical to the matched zero-pole equivalent given by Eq. (6.34). However, this is not generally true as is evident in the comparison with frequency responses of other equivalents for the third-order Butterworth filter example plotted in Fig. 6.9. Because a sample and zero-order hold is an exact model for the sample and hold with A/D converter used in the majority of discrete systems, we have already seen the computation of this equivalent in MATLAB as

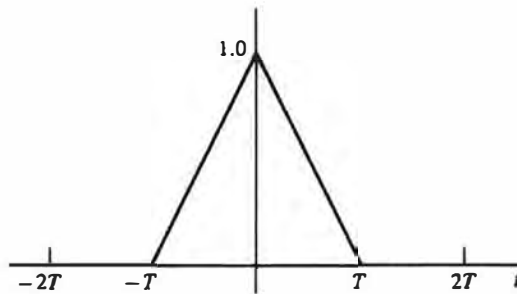
$$\text{SYSD} = \text{c2d}(\text{SYS}, T_s, \text{'zoh'})$$

where the continuous system is described by SYS and the sample period is T_s .

6.3.2 A Non-Causal First-Order-Hold Equivalent: The Triangle-Hold Equivalent

An interesting hold equivalent can be constructed by imagining that we have a noncausal first-order-hold impulse response, as sketched in Fig. 6.7. The result is called the triangle-hold equivalent to distinguish it from the causal first-order

Figure 6.7
Impulse response of the extrapolation filter for the triangle hold



hold. The effect of the triangle hold is to extrapolate the samples so as to connect sample to sample in a straight line. Although the continuous system that does this is noncausal, the resulting discrete equivalent is causal.

The Laplace transform of the extrapolation filter that follows the impulse sampling is

$$\frac{e^{Ts} - 2 + e^{-Ts}}{Ts^2}.$$

Therefore the discrete equivalent that corresponds to Eq. (6.35) is

$$H_{tri}(z) = \frac{(z-1)^2}{Tz} \mathcal{Z} \left\{ \frac{H(s)}{s^2} \right\}. \quad (6.39)$$

◆ **Example 6.4** *A Triangle-Hold Equivalent*

Compute the triangle-hold equivalent for $H(s) = 1/s^2$.

Solution. In this case, from the tables of z-transforms

$$\begin{aligned} \mathcal{Z} \left\{ \frac{H(s)}{s^2} \right\} &= \mathcal{Z} \left\{ \frac{1}{s^4} \right\} \\ &= \frac{T^3}{6} \frac{(z^2 + 4z + 1)z}{(z-1)^4}, \end{aligned} \quad (6.40)$$

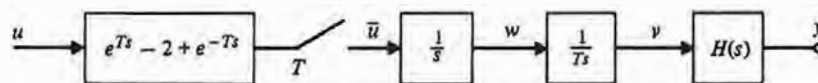
and direct substitution into Eq. (6.39) results in

$$\begin{aligned} H_{tri}(z) &= \frac{(z-1)^2}{Tz} \frac{T^3}{6} \frac{(z^2 + 4z + 1)z}{(z-1)^4} \\ &= \frac{T^2}{6} \frac{z^2 + 4z + 1}{(z-1)^2}. \end{aligned} \quad (6.41)$$

An alternative, convenient way to compute the triangle-hold equivalent is again to consider the state-space formulation. The block diagram is shown in Fig. 6.8. The continuous equations are

$$\begin{aligned} \dot{\mathbf{x}} &= \mathbf{F}\mathbf{x} + \mathbf{G}v, \\ \dot{v} &= w/T, \\ \dot{w} &= u(t+T)\delta(t+T) - 2u(t)\delta(t) + u(t-T)\delta(t-T), \end{aligned} \quad (6.42)$$

Figure 6.8
Block diagram of the triangle-hold equivalent



and, in matrix form,

$$\begin{bmatrix} \dot{\mathbf{x}} \\ \dot{v} \\ \dot{w} \end{bmatrix} = \begin{bmatrix} \mathbf{F} & \mathbf{G} & \mathbf{0} \\ 0 & 0 & 1/T \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ v \\ w \end{bmatrix} + \begin{bmatrix} \mathbf{0} \\ 0 \\ 1 \end{bmatrix} \bar{u} \quad (6.43)$$

where $\bar{u}$ represents the input impulse functions. We define the large matrix in Eq. (6.43) as F_T , and the one-step solution to this equation is

$$\zeta(kT + 1) = e^{F_T T} \zeta(kT)$$

because $\bar{u}$ consists only of impulses at the sampling instants. If we define

$$\exp(F_T T) = \begin{bmatrix} \Phi & \Gamma_1 & \Gamma_2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \quad (6.44)$$

then the equation in $\mathbf{x}$ becomes

$$\mathbf{x}(k + 1) = \Phi \mathbf{x}(k) - \Gamma_1 v(k) + \Gamma_2 w(k).$$

With care, the last two equations of Eq. (6.42) can be integrated to show that $v(k) = u(k)$ and that $w(k) = u(k + 1) - u(k)$. If a new state is defined as $\xi(k) = \mathbf{x}(k) - \Gamma_2 u(k)$, then the state equation for the triangle equivalent is

$$\begin{aligned} \xi(k + 1) &= \Phi(\xi(k) + \Gamma_2 u(k)) + (\Gamma_1 - \Gamma_2)u(k) \\ &= \Phi \xi(k) + (\Gamma_1 + \Phi \Gamma_2 - \Gamma_2)u(k). \end{aligned} \quad (6.45)$$

The output equation is

$$\begin{aligned} y(k) &= \mathbf{H}\mathbf{x}(k) + \mathbf{J}u(k) \\ &= \mathbf{H}(\xi(k) + \Gamma_2 u(k)) + \mathbf{J}u(k) \\ &= \mathbf{H}\xi(k) + (\mathbf{J} + \mathbf{H}\Gamma_2)u(k). \end{aligned} \quad (6.46)$$

Thus the triangle equivalent of a continuous system described by $[\mathbf{F}, \mathbf{G}, \mathbf{H}, \mathbf{J}]$ with sample period T is given by

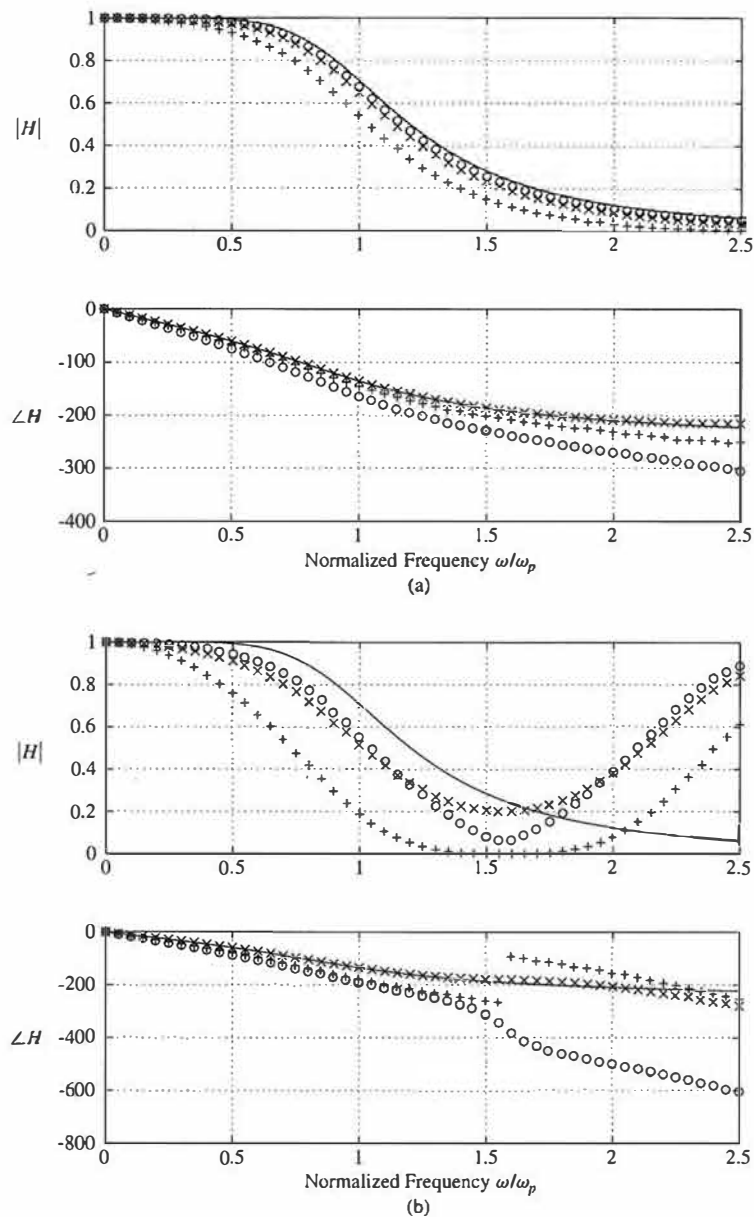
$$\begin{aligned} \mathbf{A} &= \Phi, \\ \mathbf{B} &= \Gamma_1 + \Phi \Gamma_2 - \Gamma_2, \\ \mathbf{C} &= \mathbf{H}, \\ \mathbf{D} &= \mathbf{J} + \mathbf{H}\Gamma_2, \end{aligned} \quad (6.47)$$

where Φ , Γ_1 , and Γ_2 are defined by Eq. (6.44). In the MATLAB Control Toolbox, the function `c2d` will compute the triangle-hold equivalent (referenced there as a first-order-hold equivalent) of continuous system `SYS` by

$$\text{SYSD} = \text{c2d}(\text{SYS}, T_s, \text{'foh'}).$$

Figure 6.9

Comparison of digital equivalents for sampling period (a) $T = 1$ and $\omega_s/\omega_p = 2\pi$ and (b) $T = 2$ and $\omega_s/\omega_p = \pi$ where ZOH = o, zero-pole = +, and triangle = x



In Fig. 6.9 the frequency responses of the zero-pole, the zero-order hold, and the triangle-hold equivalents are compared again for the third-order Butterworth lowpass filter. Notice in particular that the triangle hold has excellent phase responses, even with the relatively long sampling period of $T = 2$, which corresponds to a sampling frequency to passband frequency ratio of only $\omega_s/\omega_p = \pi$.

6.4 Summary

In this chapter we have presented several techniques for the construction of discrete equivalents to continuous transfer functions so that known design methods for continuous systems can be used as a basis for the design of discrete systems. The methods presented were

1. Numerical integration

- (a) Forward rectangular rule
- (b) Backward rectangular rule
- (c) Trapezoid, bilinear, or Tustin's rule
- (d) Bilinear transformation with prewarping

2. Zero-pole matching

3. Hold equivalents

- (a) Zero-order-hold equivalent
- (b) Noncausal first-order- or triangle-hold equivalent

All methods except the forward rectangular rule guarantee a stable discrete system from a stable continuous prototype with the provision that the warping frequency of the bilinear transformation with prewarping must be less than the Nyquist frequency of $\frac{\pi}{T}$ rad/sec. Zero-pole matching is the simplest method to apply computationally if the zeros and poles of the desired filter are known and takes advantage of the known relations between response and poles and zeros. This is one of the most effective methods in the context of an overall design problem and in later chapters the zero-pole matching method is frequently selected. With a reasonable computer-aided-design tool, the designer can select the method that best meets the requirements of the design. The MATLAB function `c2d` computes the discrete description for most of these discrete equivalents from a continuous system described by `SYS` with sample period T_s as follows.

$SYSD$	$=$	<code>c2d(SYS, Ts, method)</code>	where
method	$=$	'zoh'	zero-order hold
method	$=$	'foh'	first-order hold (triangle hold)
method	$=$	'tustin'	Tustin's bilinear method
method	$=$	'prewarp'	bilinear with prewarping
method	$=$	'matched'	zero-pole matching

6.5 Problems

- 6.1 Sketch the zone in the z -plane where poles corresponding to the left half of the s -plane will be mapped by the zero-pole mapping technique and the zero-order-hold technique.
- 6.2 Show that Eq. (6.15) is true.
- 6.3 The following transfer function is a lead network designed to add about 60° phase lead at $\omega_l = 3$ rad/sec

$$H(s) = \frac{s + 1}{0.1s + 1}.$$

- (a) For each of the following design methods compute and plot in the z -plane the pole and zero locations and compute the amount of phase lead given by the equivalent network at $z_l = e^{j\omega_l T}$ if $T = 0.25$ sec and the design is via
- Forward rectangular rule
 - Backward rectangular rule
 - Bilinear rule
 - Bilinear with prewarping (use ω_l as the warping frequency)
 - Zero-pole mapping
 - Zero-order-hold equivalent
 - Triangular-hold equivalent
- (b) Plot over the frequency range $\omega_l = 0.1 \rightarrow \omega_h = 100$ the amplitude and phase Bode plots for each of the above equivalents.
- 6.4 The following transfer function is a lag network designed to increase K_v by a factor of 10 and have negligible phase lag at $\omega_l = 3$ rad/sec.

$$H(s) = 10 \frac{10s + 1}{100s + 1}.$$

- (a) For each of the following design methods, compute and plot on the z -plane the zero-pole patterns of the resulting discrete equivalents and give the phase lag at $z_l = e^{j\omega_l T}$ corresponding to $\omega_l = 3$ rad/sec. Let $T = 0.25$ sec.
- Forward rectangular rule
 - Backward rectangular rule
 - Bilinear rule
 - Bilinear with prewarping (Use $\omega_l = 3$ rad/sec as the warping frequency)
 - Zero-pole matching
 - Zero-order-hold equivalent
 - Triangle-hold equivalent
- (b) For each case computed, plot the Bode amplitude and phase curves over the range $\omega_l = 0.01 \rightarrow \omega_h = 10$ rad/sec.

• 7 •

Design Using Transform Techniques

A Perspective on Design Using Transform Techniques

The idea of controlling processes that evolve in time is ubiquitous. Systems from airplanes to the national rate of unemployment, from unmanned space vehicles to human blood pressure, are considered fair targets for control. Over a period of three decades from about 1930 until 1960, a body of control theory was developed based on electronic feedback amplifier design modified for servomechanism problems. This theory was coupled with electronic technology suitable for implementing the required dynamic compensators to give a set of approaches to solve control problems now often called **classical techniques** to distinguish these methods from designs based on a state-space formulation which came to be called **modern techniques**. The landmark contributors to this “classical” theory are Evans (1950) [root locus] and Nyquist (1932) and Bode (1945) [frequency response]. For random inputs, the work of Wiener (1948) should be added. The unifying theme of these methods is the use of Fourier and Laplace transforms to represent signals and system dynamics and to describe the control specifications. Controller design is then carried out in the selected transform domain. From the perspective of the 90’s the terms “classical” and “modern” seem a bit pejorative and we prefer to classify the methods as **transform techniques** and **state-space techniques**.

The methods based on transforms were developed before computers were available and the engineer had to depend on hand calculations and careful hand plotting to achieve the design. The availability of computers and software such as MATLAB have made calculations and plotting simple, fast and accurate; and now the hand-plotting guidelines are used as verification of the automatic calculations and as a guide to design decisions. In this role, the understanding of the design process gained by the experience of doing a simple design by hand is well

worth the effort spent in developing the required skills. The introduction of digital control and sampled data adds new constraints and new possibilities to the transform design methods. The z -transform is added to the Laplace and the Fourier transforms and poles and zeros have meaning relative to the unit circle rather than to the imaginary axis. The meaningful part of the frequency response is restricted to half the sampling frequency. Each of these developments must be understood in order to apply transform methods to digital control.

Chapter Overview

design by emulation

Building on previous understanding of the design of continuous systems, the first method for digital design is based on **emulation** of a continuous design. The continuous controller is simply replaced with a digital equivalent computed by using one of the techniques described in Chapter 6. The result may be evaluated in terms of poles and zeros in the z -plane, magnitude and phase in the frequency response, or transient response to step, impulse or other input.

design by root locus

The second method introduced is the **root locus** where it is demonstrated that the rules of the root locus are unchanged from the continuous case but the relations between pole location and time response must refer to the z -plane rather than the s -plane.

design by frequency response

Finally, the Nyquist stability criterion for discrete systems is developed and Bode's design methods for gain and phase margins are extended to discrete systems. In addition to the usual results, the concept of system sensitivity is developed to show how **frequency response** can be used to cause the system to be robust with respect to both stability and performance when the plant transfer function is subjected to bounded but unknown perturbations.

7.1 System Specifications

We first consider the design specifications that the controller is expected to achieve. As reviewed in Chapter 2, the central concerns of controller design are for good transient and steady-state response and for sufficient robustness. Requirements on time response and robustness need to be expressed as constraints on s -plane pole and zero locations or on the shape of the frequency response in order to permit design in the transform domains. Dynamic performance in the time domain is defined in terms of parameters of system response to a step in command input. The most frequently used parameters are the rise time, t_r ; the settling time, t_s ; the percent overshoot, M_p ; and the steady-state error, e_{ss} . These parameters, which apply equally well to discrete control as to continuous control, are discussed in Section 4.1.7. The s -plane expressions of these requirements are summarized by the following guidelines:

- The requirement on natural frequency is

$$\omega_n \geq 1.8/t_r. \quad (7.1)$$

- The requirement on the magnitude of the real part of the pole is

$$|\operatorname{Re}\{s_i\}| = \sigma = \zeta\omega_n \geq 4.6/t_s. \quad (7.2)$$

- The fractional overshoot, M_{pr} , is given in terms of the damping ratio, ζ , by the plot of Fig. 2.7 which can be very crudely approximated by

$$\zeta \approx 0.6(1 - M_p). \quad (7.3)$$

The specifications on steady-state error to polynomial inputs is determined by the error constant appropriate for the case at hand as described in Section 4.2.2. The most common case is for systems of Type 1, which is to say, systems that have zero steady-state error to a step input and finite error to a ramp input of slope r_0 of size $e_{ss} = r_0/K_v$ where K_v is the velocity constant. For a single-loop system with unity feedback gain and forward transfer function $D(s)G(s)$ as shown in Fig. 7.1, the system is **Type 1** if DG has a simple pole at $s = 0$.

velocity constant

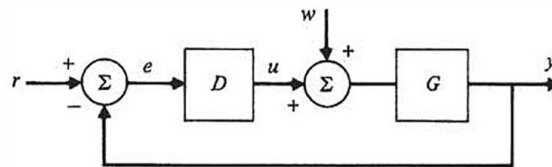
- The **velocity constant** is then given by

$$K_v = \frac{r_0}{e_{ss}} = \lim_{s \rightarrow 0} s D(s)G(s).$$

The fact that in discrete systems the control is applied as a piecewise constant signal causes a roughness in the response that is directly related to the sample frequency. A specification on roughness leads to a specification on sample period, T . This parameter is so important to the design of discrete controls that Chapter 11 is devoted to the decision. At this point, let it suffice to point out that the smaller the T , the better the approximation to continuous control and the smaller the roughness.

- A reasonable choice of T is one that results in at least 6 samples in the closed-loop rise time and better, smoother control results if there are more than 10 samples in the rise time.

Figure 7.1
A unity feedback system



◆ **Example 7.1** *Selection of Sample Period*

What is the relation between sampling frequency and system natural frequency if there are 10 samples in a rise time?

Solution. The sampling frequency in radians/sec is given by $\omega_s = 2\pi/T$ and we assume that rise time and natural frequency are related by Eq. (7.1) so that

$$\begin{aligned}\omega_n &= 1.8/t_r \\ &= \frac{1.8}{10T}\end{aligned}$$

Substituting for T , we find that

$$\omega_n = \frac{0.18\omega_s}{2\pi}$$

or

$$\omega_n \approx \omega_s/35.$$

In other words, the sample rate, ω_s , should be 35 times faster than the natural frequency, ω_n .

From this example, we conclude that typically the sample frequency should be chosen to be 20 to 35 times the closed loop natural frequency. Slower sampling can be used but one would expect the resulting transients to be excessively rough.

Robustness is the property that the dynamic response (including stability of course) is satisfactory not only for the nominal plant transfer function used for the design but also for the entire class of transfer functions that express the uncertainty of the designer about the dynamic environment in which the real controller is expected to operate. A more comprehensive discussion of robustness will be given when design using frequency response is considered. For root locus design, the natural measure of robustness is, in effect, gain margin. One can readily compare the system gain at the desired operating point and at the point(s) of onset of instability to determine how much gain change is acceptable.

- A typical robustness requirement is that one should have gain margin of two so that the loop gain must double from the design value before reaching the stability boundary.

7.2 Design by Emulation

The elements of design by emulation have been covered already. Continuous control design is reviewed in Chapter 2, and in Chapter 6 the techniques of computing discrete equivalents are described. Control design by emulation is

mainly a combination of these two ideas. A controller design is done as if the system is to be continuous and, after a sample period is selected, a discrete equivalent is computed and used in place of the continuous design. This discrete controller may then be simulated and tested in the discrete control loop and modifications made, if necessary.

7.2.1 Discrete Equivalent Controllers

Techniques to compute discrete equivalents are described in general terms in Chapter 6, and their performance is illustrated on the basis of filter frequency responses. In this chapter, we are interested in controllers for feedback control and in performance comparisons on the basis of time responses. Any of the techniques from Chapter 6 can be used for the purpose; here we illustrate the use of the pole-zero mapping equivalent and explore the choice of sample period by example. An alternative approach that considers directly the performance for the discrete controller in the feedback context has been described by Anderson (1992). The method described in that reference leads to a multirate sampling problem of the sort which will be considered in Chapter 11.

◆ Example 7.2 *Design of Antenna Servo Controller*

A block diagram of the plant for an antenna angle-tracker is drawn in Fig. 7.2. The transfer function is given by

$$G(s) = \frac{1}{s(10s + 1)}.$$

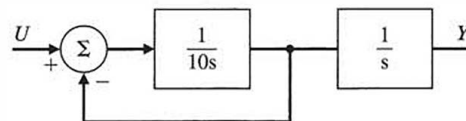
The specifications for this system are

1. Overshoot to a step input less than 16%
2. Settling time to 1% to be less than 10 sec
3. Tracking error to a ramp input of slope 0.01 rad/sec to be less than 0.01 rad
4. Sampling time to give at least 10 samples in a rise-time

Design a controller for this system using the method of emulation.

Solution. From the specifications one can estimate the acceptable region in the s -plane for the closed loop poles. From the overshoot requirement, we conclude that the damping ratio

Figure 7.2
Block diagram of the
plant transfer function



must be $\zeta \geq 0.5$. From the settling time requirement, we conclude that the roots must have a real part of $\sigma \geq 4.6/10 = 0.46$. Finally, from the steady-state error requirement, we conclude that the velocity constant is constrained to be $K_v \geq \frac{0.01}{0.01} = 1.0$. Based on the limits on the damping ratio and the real-part of the poles, we can sketch the acceptable region for closed-loop poles in the s -plane as done in Fig. 7.3. Using lead compensation to cancel a plant pole, a first choice for controller might be

$$D(s) = \frac{10s + 1}{s + 1}. \quad (7.4)$$

The root locus for this choice is drawn in Fig. 7.4 using MATLAB commands to enter the plant as a system, `ant`, the compensation as `lead1`, and the product as the open loop system, `sysol`.

```
np = 1;
dp = [10 1 0];
ant = tf(np,dp);
nc = [10 1];
dc = [1 1];
lead1 = tf(nc,dc);
sysol = lead1*ant;
rlocus(sysol)
```

Figure 7.3
Acceptable pole
locations for the
antenna control

